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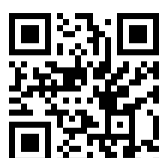
FINANCIAL STABILITY

Regulating and supervising digital assets: where does control lie?

Centralisation and decentralisation across issuer, protocol and access layers

The EU has established, through the Markets in Crypto-Assets Regulation (MiCAR), a broad but residual framework for crypto-assets, including specific rules for stablecoin issuers and a harmonised regime for crypto-asset service providers (CASPs). As this framework moves from rule-making to implementation and supervision, a key question is whether regulators and supervisors can identify where control, risk and responsibility actually lie in increasingly complex digital-asset arrangements. Legal classification alone may not provide a full answer: an identifiable issuer may make a legal promise, a public protocol may distribute validation and governance, while exchanges, custodians, wallets and interfaces may concentrate practical control.

This briefing examines which characteristics of digital-asset arrangements determine where effective regulation and supervision can act. It develops a layer-based approach covering the instrument, protocol and access layers and applies it to corporate stablecoins and Ethereum. The analysis suggests that policymakers should assess the location of control, transparency, legal and jurisdictional reach, and the practical capacity to reduce risk, rather than treating centralisation or decentralisation as labels for an arrangement as a whole.



1. Introduction: from legal classification to the location of control

The EU has established through the [Markets in Crypto-Assets Regulation](#)¹ (MiCAR) a harmonised, but residual, regulatory framework for crypto-assets not already covered by existing EU financial legislation. MiCAR differentiates among asset-referenced tokens (ARTs), e-money tokens (EMTs) and other crypto-assets within its scope, while establishing a separate regime for crypto-asset service providers (CASPs). Together with the [Digital Operational Resilience Act](#)² (DORA), the EU AML/CFT framework and other sectoral legislation, it provides regulators and supervisors with a broad set of tools addressing prudential, conduct, operational and financial-integrity risks. As these rules are implemented in practice, however, an additional supervisory question arises: where, within a digital-asset arrangement, are the functions that create risk and the actors or mechanisms capable of controlling them?

CONTENT

1. [Introduction: from legal classification to the location of control](#)
2. [What centralisation and decentralisation mean in regulatory terms](#)
3. [Corporate stablecoins: digital money issued by private firms](#)
4. [Decentralised networks: Ethereum as infrastructure](#)
5. [AML and enforcement across issuer, protocol and access layers](#)
6. [Consumer protection and market integrity](#)
7. [Financial stability and monetary sovereignty](#)
8. [Concluding observations and key policy](#)

This question matters because digital-asset arrangements do not necessarily correspond to a single organisational model. An identifiable issuer may create a token and owe holders a redemption or other legal claim, while the token circulates on a public distributed network. Conversely, a network whose transaction validation is widely distributed may depend in practice on a limited number of staking providers, infrastructure services, exchanges, custodians or user interfaces. **Legal classification remains essential, but it does not by itself show where operational dependencies arise, who can change relevant rules, who can intervene when problems occur, or whether the actors exercising practical control are transparent and within effective legal and supervisory reach.**

Against this background, **the research question of this briefing is which characteristics of a digital-asset arrangement determine the appropriate points of regulatory and supervisory intervention.** More specifically, it asks how the distribution of control across different layers affects responsibility, legal reach and the capacity to prevent, mitigate or remedy risk. The analysis does not seek to determine whether centralised or decentralised models are inherently preferable, nor does it assume that decentralisation necessarily reduces supervisability. Instead, it treats centralisation and decentralisation as characteristics of particular functions within an arrangement.

Methodologically, the briefing approaches the issue from the characteristics of the arrangement rather than from legal categories alone, and it distinguishes three layers: a) at the **instrument layer**, the relevant questions concern what the token represents, whether an issuer exists and what legal promise or claim attaches to the asset; b) at the **protocol layer**, the analysis considers transaction validation, software rules and governance, including who can materially influence or change those rules; and c) at the **access layer**, it examines the exchanges, custodians, wallets, interfaces and infrastructure through which users enter, use and leave the arrangement. Across these layers, the assessment focuses on four elements: the allocation of

¹ Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets.

² Regulation (EU) 2022/2554 of the European Parliament and of the Council of 14 December 2022 on digital operational resilience for the financial sector.

control, transparency, legal and jurisdictional reach, and the practical capacity of actors or technical mechanisms to reduce risk.

The briefing applies this framework to two case studies located at different layers: corporate stablecoins and Ethereum. Corporate stablecoins illustrate an instrument with an identifiable issuer, off-chain reserves and a redemption commitment; Ethereum illustrates a base-layer network in which validation and protocol governance are distributed across multiple participants. They are not presented as mutually exclusive alternatives, given that corporate stablecoins can, and frequently do, circulate on public networks such as Ethereum. Their comparison is therefore intended to show how different forms of control and dependency can coexist within the same digital-asset ecosystem and how their location affects the regulatory tools available.

The subsequent analysis considers the implications of this layered approach for AML/CFT and enforcement; consumer protection and market integrity; and financial stability and monetary sovereignty. **The purpose is not to propose a new legal classification of digital assets, but to complement existing regulatory categories with a practical framework for identifying where supervision can operate effectively**, where responsibility can be assigned and where concentrations or dependencies may require closer monitoring.

2. What centralisation and decentralisation mean in regulatory terms

For a regulatory analysis, centralisation should be treated as a property of a specific function rather than of the arrangement as a whole. Issuance and redemption may be controlled by one entity, transaction validation may be distributed, protocol upgrades may depend on a limited group, and user access may be concentrated in a small number of intermediaries. **Decentralisation therefore describes the distribution of a particular power or dependency: who can perform the function, who can alter it, and whether one actor can exercise decisive control.** This approach avoids equating identifiability with supervisability. An identifiable issuer may be difficult to supervise if it is opaque or outside the relevant jurisdiction, while a public ledger may make activity observable and allow some controls to be automated. At the same time, transparent code does not by itself create legal responsibility, remedy losses or ensure that governance participants are within regulatory reach. As said, the assessment should therefore combine **four elements: the allocation of control, transparency, legal and jurisdictional reach, and the practical ability of actors or technical mechanisms to reduce risk.** In this briefing, centralisation refers to the allocation of control within a digital-asset arrangement, and should not be confused with the centralisation of public supervisory competences, which raises a separate set of institutional trade-offs concerning consistency, coordination, expertise and accountability.

For regulatory purposes, the most useful questions concern the location and nature of dependency. Who creates or sponsors the token, and who owes any claim attached to it? Who can change the protocol or smart-contract rules? Who holds reserves, administrative keys or user assets? Who can stop, reverse or freeze a transaction? Which actors or governance arrangements are within effective legal and supervisory reach? Are there intermediaries or infrastructure providers without which most users cannot enter, use or leave the arrangement? These questions identify where regulation, supervision, contractual safeguards or code-based controls may operate.

A practical assessment of centralisation should begin by identifying the actors that exercise real control. Box 1 presents six basic questions that can help legislators and supervisors locate responsibility and dependency within a digital asset arrangement.

Box 1: Key questions for locating control across layers**Where is control and who can be reached?**

When analysing a digital-asset arrangement, six questions should be considered:

1. At the instrument layer, who creates or sponsors the token, and who owes any claim attached to it?
2. At the protocol layer, who can propose, approve or implement changes to the rules?
3. Who holds reserves, administrative keys or user assets?
4. Who can stop, reverse, freeze or otherwise influence transactions?
5. Which actors or governance processes are transparent and within effective legal or supervisory reach?
6. Which intermediaries or infrastructure providers are necessary for most users to access or exit the arrangement?

Source: EGOV elaboration.

Regulatory action does not need to focus only on the base protocol. Figure 1 shows the different layers of a digital asset ecosystem and identifies the points at which public authorities can most effectively apply regulatory requirements.

Figure 1: Where regulation can act: protocol vs ecosystem

Where regulation can act: protocol vs ecosystem				
Ecosystem layer	Example actors	Concentration	Identifiability	Potential regulatory leverage
Stablecoin issuer	Circle, Tether	High	Few identifiable issuers	High
Exchange	Binance, Coinbase, Kraken	Medium-high	Identifiable firms	High
Custodian	Coinbase Custody, BitGo, Fireblocks	Medium	Identifiable firms	High
Staking provider	Lido, Coinbase, Binance	High	Few large providers	Medium-high
Wallet / interface	MetaMask, Phantom, Coinbase Wallet	Medium	Some identifiable interfaces	Medium
Validator operator	Ethereum validators / node operators	Low-medium	Distributed / partially identifiable	Low

Sources: EGOV elaboration based on MiCAR, ESMA and Ethereum.org.

Note: Regulatory leverage depends on identifiability, legal and jurisdictional reach, and the actor's technical ability to influence the relevant function. A protocol or governance arrangement may also accept enforceable obligations. The assessment is illustrative and does not represent a formal legal classification.

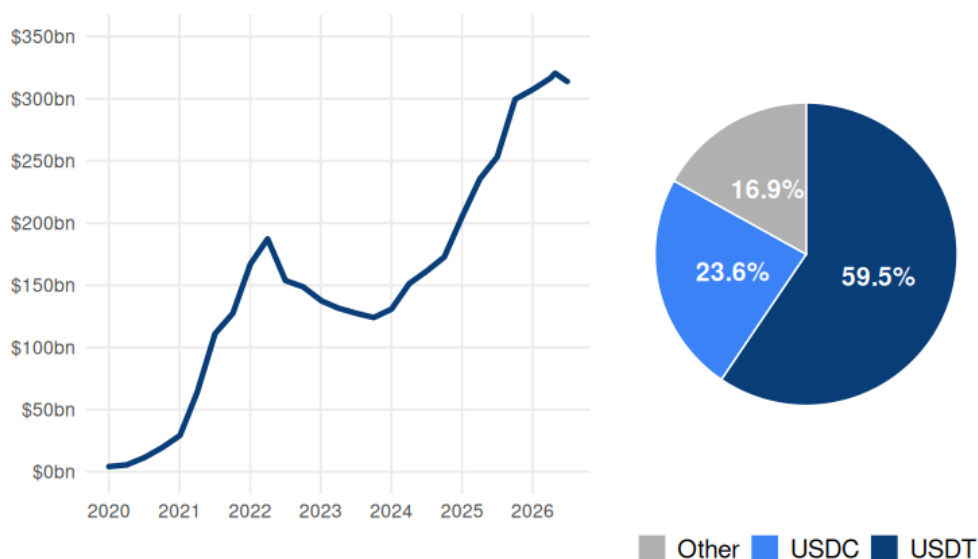
Another useful question is what type of trust the arrangement requires from users. **In a corporate stablecoin, users rely on the issuer's legal promise and reserve management, as well as on banks, custodians and auditors, while in a decentralised protocol, users rely more on software, validation, governance procedures and the continued acceptance of the rules by network participants.** These forms of trust can coexist and can fail in different ways. Where an identifiable issuer or service provider is within effective jurisdictional reach, the law can impose duties concerning authorisation, own funds, reserves, conflicts of interest, complaints, redemption and disclosure, but where no such issuer exists, regulation may focus on intermediaries and other actors with practical control. Where a protocol or governance

arrangement has a sufficient legal and jurisdictional connection, the law may also set conditions for material rule changes or require governance procedures that make regulatory approval possible. The appropriate method therefore depends not only on whether an issuer exists, but also on legal reach, the allocation of control and the feasibility of enforcing or embedding safeguards.

3. Corporate stablecoins: digital money issued by private firms

This section focuses on corporate stablecoins such as USDC (USD Coin) and USDT (Tether): tokens issued by identifiable firms that seek to maintain a stable value through reserve assets and a redemption arrangement. Corporate stablecoins are widely used in crypto markets because they allow users to move value without leaving the blockchain environment. They can also be used as settlement assets, collateral in trading, and sometimes as a payment tool. **Even though their name suggests stability, this is not automatic, it depends on the design of the stablecoin and on the quality of the assets that support it.** Thus, in a corporate stablecoin model, a private issuer creates tokens and promises that users can redeem them at a fixed value, so that if a token is meant to be worth one dollar, the issuer must be able to return one dollar, or an equivalent value, when the user asks for redemption, and to back up this promise the issuer holds reserves. These reserves may include bank deposits, short-term government securities or other liquid assets, according to the design and the law that applies, with the token's stability relying on off-chain assets. Ultimately, the blockchain records the token, but the trust comes from reserve management and legal rights. This structure is indeed familiar for regulators, as it has some similarities with electronic money, payment institutions and money market instruments. It is not identical to any one of them, but it raises similar questions: Are the reserves safe and liquid? Are they separated from the issuer's own assets? Can users redeem quickly? Who audits the reserves? What happens if many users redeem at the same time? What happens if a bank holding the reserves fails? The stablecoin market has grown rapidly, but it remains highly concentrated. Figure 2 shows the quarterly average of total stablecoin market capitalisation from 2020 to the second quarter of 2026, together with token-level market shares at the end of the period.

Figure 2: Stablecoins: growth and concentration (Panel A – Total market capitalisation, quarterly averages, 2020–Q2 2026; Panel B – Market shares by token, Q2 2026)



Sources: DefiLlama Stablecoins Dashboard (total market capitalisation time series) and DefiLlama Stablecoins API (token-level market capitalisation).

The main corporate stablecoins follow a similar centralised model, but they differ in their issuer, legal location, reserve arrangements and disclosure practices. Figure 3 identifies the main entities behind these tokens and the principal ways in which they exercise control.

Figure 3: Main stablecoins: issuer, jurisdiction, reserves and disclosure

Who controls the main stablecoins?				
Token	Issuer	Jurisdiction	Reserve type	Disclosure
USDT	Tether	El Salvador	Treasuries, repos, cash, gold and other assets	Daily transparency + quarterly attestation
USDC	Circle	United States	Cash and short-term U.S. Treasuries	Weekly reserve reports + monthly attestation
EURC	Circle	France (MiCA)	Euro-denominated reserve assets	Monthly attestation
EURCV	Societe Generale-FORGE	France	Cash reserves	Daily disclosure + monthly attestation
EURI	Banking Circle	Luxembourg	Euro cash deposits	Issuer disclosures and attestations

Sources: DefiLlama Stablecoins API; Circle and Tether transparency reports; MiCAR documentation.

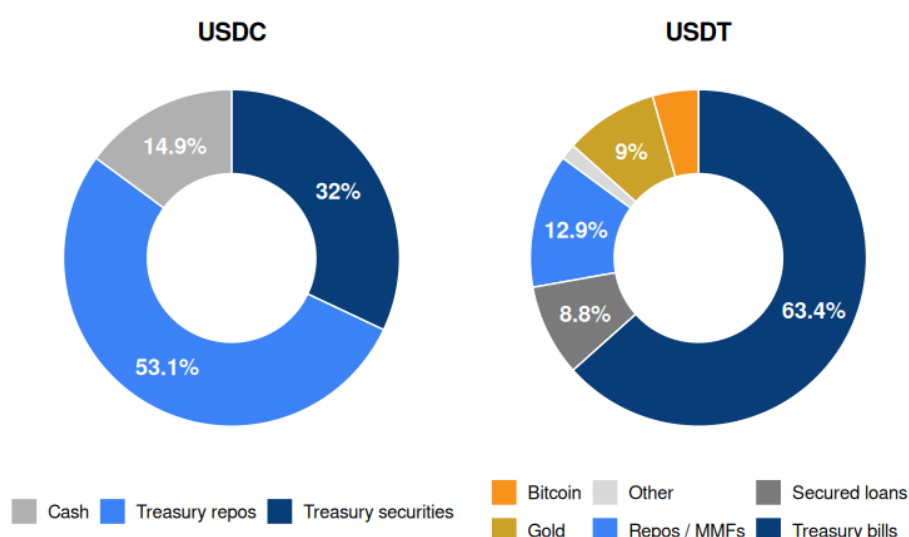
The fact that corporate stablecoins have a clear centre of trust can be useful for users too, as a regulated issuer can establish a customer service, be the recipient of a legal claim, and count with internal controls and compliance procedures. But the token is not a neutral infrastructure as users depend on the issuer's decisions, technology and business continuity. This dependence becomes more important when a stablecoin grows, because **if a stablecoin is used mainly inside crypto trading, the risks may stay mostly inside the crypto market, but if it becomes widely used for payments, savings or settlement, the risks can move closer to the real economy and the banking system.** In this regard, the [Financial Stability Board](#) (FSB) has developed recommendations for global stablecoin arrangements calling for comprehensive regulation, supervision and oversight, including cross-border cooperation, because a large stablecoin may operate across jurisdictions and sectors. A corporate stablecoin is not only a token recorded on a blockchain; it is also a private monetary arrangement that may offer faster transfer and settlement while raising questions about trust, liquidity, redemption and governance. The use of distributed ledger technology does not remove the need for effective oversight of the issuer and the wider arrangement. Table 1 summarises the basic governance structure of a corporate stablecoin. It links each main feature of the model to its most important regulatory consequence.

Table 1: Corporate stablecoins: governance and policy implications

Question	Corporate stablecoin answer	Policy meaning
Who creates the token?	A private issuer	The issuer can be authorised and supervised where it is within effective jurisdictional reach
Why is the token stable?	It is backed by reserves and a redemption promise	Reserve quality and redemption rights are central
Who can act in a crisis?	The issuer and public authorities	There is a clear point of contact, but also a single point of failure

Source: EGOV elaboration.

The credibility of a corporate stablecoin depends largely on the quality and liquidity of the assets that support it. Figure 4 compares the composition of the reserves and the disclosure practices of two major stablecoin issuers.

Figure 4: What backs a stablecoin? (Reserve composition and disclosure)

Sources: Circle Reserve Fund Report (November 2025) and Tether Financial Figures and Reserves Report (Q4 2025).

Note: Reserve categories are based on issuer disclosures and are not fully harmonised. The figures should therefore be read as an indicative comparison.

3.1. Regulatory advantages of identifiable and reachable actors

First, **an identifiable actor can make legal accountability more direct where it is within effective jurisdictional reach**. A company that issues a stablecoin can be required to obtain authorisation, meet capital and governance standards, report information, protect users and manage risks. It can also be contacted, required to provide information and, where appropriate, sanctioned. This advantage comes from identifiability and legal reach rather than from centralisation alone: an opaque offshore issuer may be difficult to supervise, while a public protocol may provide transparent data and support automated controls. Anti-money laundering (AML) requirements may also be applied through the issuer and service providers, which can conduct customer due diligence, monitor transactions, cooperate with authorities and, where legally

required and technically possible, freeze assets. The Financial Action Task Force (FATF) has underlined that virtual assets remain vulnerable to illicit finance when standards are not applied consistently across jurisdictions, and its [2025 update](#) highlights the growing use of stablecoins by illicit actors. This does not mean that stablecoins are mainly used for illicit purposes; it means that large and fast-moving digital payment instruments require effective and internationally coordinated controls.

Second, **for consumers, a regulated centralised model can create clearer rights, as users may receive information before buying or using a token, know who the issuer is, how redemption works, what fees apply, and where to complain.** In a decentralised network, the user may have more direct control, but also more personal responsibility. If, for example, a user loses a private key, sends assets to the wrong address or uses a risky smart contract, there may be no central entity that can reverse the transaction. Centralised models can also help market confidence, given that clear disclosure about reserves, audits and governance can reduce rumours and panic. Additionally, in case of a stress event, supervisors can require information and coordinate with other authorities and the issuer can communicate with users and manage redemption procedures. This does not guarantee stability, but it gives public authorities tools that would not be possible without a visible operator.

The third advantage is the integration with the existing financial law. **Stablecoins that function like money can be connected to payment law, e-money rules, prudential requirements and conduct rules.** MiCAR is a good example of this functional approach. The [European Securities and Markets Authority](#) (ESMA) describes MiCAR as a mechanism to create uniform market rules in the EU for crypto-assets, including transparency, disclosure, authorisation and supervision for issuers and service providers, an approach that seeks to avoid a gap between traditional finance and digital assets.

Fourth, **centralised models may also be easier for institutions to use, given that banks, payment firms and regulated investors need clear legal relationships and identifiable counterparties,** and a corporate stablecoin issuer can sign contracts, provide documentation and meet regulatory reporting needs, while a decentralised protocol may be harder for such firms to assess, even if the technology is robust. For this reason, corporate stablecoins may be more likely to enter mainstream payment and settlement channels.

However, these advantages should not lead to the conclusion that centralisation is always safer, it can improve accountability only where the central actor is well governed, effectively supervised and able to meet its obligations. If the actor is weak, opaque, outside effective supervisory reach or so significant that authorities cannot let it fail safely, concentration can increase systemic risk. The policy challenge is to obtain the benefits of clear responsibility without allowing private issuers to become unchecked gatekeepers of digital money.

3.2. Risks created by corporate stablecoins

The main risk in a corporate stablecoin is that the promise of stability may break, and this can happen if users doubt the quality of the reserves, if the issuer cannot redeem quickly, if a custodian or bank fails, or if the issuer faces legal or operational problems. A stablecoin can then suffer a run, meaning many users attempt to redeem their cryptocurrencies simultaneously, forcing the issuer to sell assets rapidly and potentially causing losses or putting pressure on other markets. Therefore, the risk is not limited to the blockchain itself, as it can connect with banks and securities markets through the reserve assets.

A second risk is concentration: when a small number of stablecoins dominate digital markets, they become important gateways, and trading platforms, decentralised finance applications and users may all depend on them, which can create a single point of failure, so that a technical outage, legal issue, cyber incident or reserve problem at one major issuer could affect many markets simultaneously. In this regard, the [European Systemic Risk Board](#) (ESRB) warned in 2025 that financial stability risks from crypto-assets, including stablecoins, were rising as links with the financial sector increased.

A third risk is concentrated control over access and transfer. A corporate stablecoin issuer may be able to freeze addresses, blacklist users or restrict transfers. These tools can support legitimate public objectives, such as sanctions enforcement or asset recovery, but they also place significant power in the issuer and, indirectly, in the jurisdiction whose legal orders the issuer must follow. Users may not always understand this allocation of control: they may believe they are using open blockchain-based money while remaining subject to the issuer's legal and technical restrictions. In arrangements where no actor has comparable administrative powers, intervention may instead depend on intermediaries, protocol governance or action at access points.

A fourth risk is regulatory fragmentation. Stablecoins are cross-border by design, so a token issued in one country may be used by people in many other countries. If the issuer, reserve assets, users and trading venues are located in different jurisdictions, supervisors may not have the same powers or information. The [FSB recommendations](#) for global stablecoins stress the need for authorities to cooperate and to supervise arrangements across borders and functions. Without this cooperation, risks can move to the weakest jurisdiction and then return to the rest of the system.

A fifth risk concerns monetary sovereignty. If foreign-currency stablecoins become widely used for payments or savings, they may increase the role of that currency in the domestic economy. This issue is examined further in Section 7.

A sixth risk concerns the specific operational dependencies created by the corporate stablecoin model. Like other digital assets, a stablecoin relies on software, cybersecurity and communications infrastructure. In addition, its value and redeemability depend on the continuous operation of the issuer, reserve-management systems, banks, custodians and payment channels that connect the on-chain token to off-chain assets. A distributed ledger may provide redundancy in transaction recording, but it does not by itself remove these issuer-side and reserve-side dependencies; distributed infrastructure can also be used in more centralised arrangements. Supervision should therefore assess the whole service chain, including governance, outsourcing, cyber resilience, reserve access and the capacity to continue issuance, transfers and redemption during stress.

A final risk is user misunderstanding. The term *stablecoin* may suggest a level of safety similar to a bank deposit or central bank money, although the legal rights and protection arrangements are different. Section 6 examines the related consumer-protection requirements.

The EU addresses many of the risks created by corporate stablecoins through MiCAR. Box 2 summarises how the Regulation classifies stablecoins, places duties on issuers and service providers, and defines the main limits of the EU regulatory perimeter.

Box 2: The EU response: MiCAR and the regulatory perimeter**MiCAR and the EU regulatory perimeter**

Under [MiCAR](#), the EU distinguishes between two main categories of stablecoins, asset-referenced tokens³ (ARTs) and e-money tokens⁴ (EMTs). In simple terms, an EMT refers to one official currency, while an ART refers to another value, right or combination of values⁵. MiCAR requires issuers of these tokens to meet operating and authorisation requirements to operate in the EU, while [EBA technical standards and guidelines](#) add detail to the framework, covering areas such as liquidity management, own funds, conflicts of interest, stress testing and other prudential matters. **When an ART or EMT is classified as significant, the EBA assumes enhanced supervisory responsibilities at EU level.** This shows that the EU treats stablecoins not only as technology, but as financial arrangements that need governance, transparency and prudential safeguards.

MiCAR does not treat all crypto-assets alike. It distinguishes ARTs, EMTs and a residual category of other crypto-assets, and applies separate requirements to issuers and CASPs according to the function performed. **Together with EBA standards, DORA and other EU financial legislation, it addresses many governance, disclosure, prudential, conduct and operational-resilience risks.** The paper therefore does not identify a general failure by MiCAR to differentiate among token types.

The remaining questions arise mainly at the edges of this actor-based framework. They include arrangements with no identifiable issuer or operator; issuers, services or access points outside effective EU jurisdiction; and concentrations of control in protocol governance, staking, cloud, node or interface providers that are not captured in the same way as issuers and regulated CASPs. Mixed arrangements may also combine an EU-regulated issuer with a public network and third-country access points. These issues call for careful perimeter assessment, coordination with DORA and other sectoral rules, effective third-country market-access conditions and international supervisory cooperation; they do not necessarily require a new MiCAR token category.

Technology-neutral regulation should therefore remain sensitive to the layer at which the risk arises. For corporate stablecoins, the central issues are the issuer's promise, reserves and redemption. For decentralised protocols, they are governance, validation, code and concentration. For users, the relevant risks also include the intermediaries through which the arrangement is accessed.

Source: EGOV elaboration.

4. Decentralised networks: Ethereum as infrastructure

[Ethereum](#) is examined here as a base-layer network rather than as the opposite of a stablecoin. Stablecoins and other tokens may be issued and transferred on Ethereum, while Ether (ETH) is used within the network to support its operation. The two examples therefore sit at different layers: Section 3 concerns the issuer, reserve and redemption arrangements of an instrument, whereas this section concerns validation, protocol rules and infrastructure. Ethereum uses [proof-of-stake](#), under which validators lock ETH and help verify transactions and blocks; dishonest behaviour can lead to the loss of part of the stake. **Network security therefore relies on software, economic incentives and distributed participation rather than on an issuer's reserve assets, and changes to the protocol require coordination among developers, validators, node operators, applications and users.** This can limit unilateral change, but it does not eliminate power, because some developers, foundations, infrastructure providers or large validators may exercise greater influence than ordinary users.

From a policy point of view, this model offers resilience, transparency and opportunities for automated controls. If no single company runs the network, the failure of one company should not automatically paralyse the protocol, and public rules and transaction records may also make some forms of monitoring easier, and safeguards can be embedded in smart contracts or governance procedures. Users may choose

³ Defined as "a type of crypto-asset that is not an electronic money token and that purports to maintain a stable value by referencing another value or right or a combination thereof, including one or more official currencies" by Article 3(1)(6) of Regulation (EU) 2023/1114.

⁴ Defined as "a type of crypto-asset that purports to maintain a stable value by referencing the value of one official currency" by Article 3(1)(7) of Regulation (EU) 2023/1114.

⁵ MiCAR also covers a residual category of crypto-assets that are neither asset-referenced tokens nor e-money tokens (Title II). Although much of the regulatory and supervisory discussion has focused on stablecoins, particularly ARTs and EMTs, the Regulation also establishes requirements for the public offering and admission to trading of other crypto-assets falling within its scope. Additionally, MiCAR explicitly excludes from its scope crypto-assets already covered by existing EU sectoral financial legislation.

different wallets, nodes or service providers, and developers can build applications without permission from a central operator. **The main regulatory challenge is that responsibility and legal reach may be distributed or uncertain, making it more difficult to identify where conventional regulatory obligations can operate effectively.** If users lose money because of a protocol issue, smart-contract error or manipulation around a decentralised service, it may be unclear which actor has a legal duty or the practical capacity to provide a remedy. **Consequently, authorities may regulate identifiable gateways and service providers** and, where a governance arrangement has a sufficient jurisdictional link, may also apply requirements to governance or rule-change processes, and the question is which intervention is effective and proportionate at each layer.

4.1. Practical centralisation inside decentralised systems

Decentralisation should be assessed in practice, not only in theory, because a network may be decentralised at the protocol level while important functions remain concentrated in the way it is governed, operated or accessed. In practice, concentration can arise at several points: validation and protocol governance, supporting infrastructure, and the exchanges, custodians and interfaces through which most users interact with the network.

At the protocol layer, validation may become concentrated even where participation is formally open. Many users do not participate directly in validation and instead rely on staking providers or pools. If a small number of providers control a large share of validation, they may acquire significant influence over network operations and may also become important points of operational, regulatory or cyber risk. Proof-of-stake is designed to support broad participation, but economic incentives can still favour larger providers because users may consider them easier, cheaper or safer. This can create a gap between the decentralised design of the protocol and the actual structure of participation.

Protocol governance may also be influenced by a relatively limited group of developers or other technically important participants. Open-source software can be reviewed and copied by many people, which supports transparency, but in practice only a limited number of developers may have the expertise and reputation needed to propose or implement major changes. If users, validators and node operators normally follow software maintained by a core group, that group may exercise significant influence over the evolution of the network. This is not necessarily equivalent to the legal control exercised by a corporate issuer, but it is relevant when assessing where practical influence lies.

Infrastructure can create another form of concentration. Many applications rely on the same node-service providers, cloud providers, data services or developer tools. If one of these services fails, users may lose access to applications even though the underlying protocol continues to operate. A decentralised protocol can therefore remain technically resilient while the infrastructure supporting its practical use is concentrated or fragile.

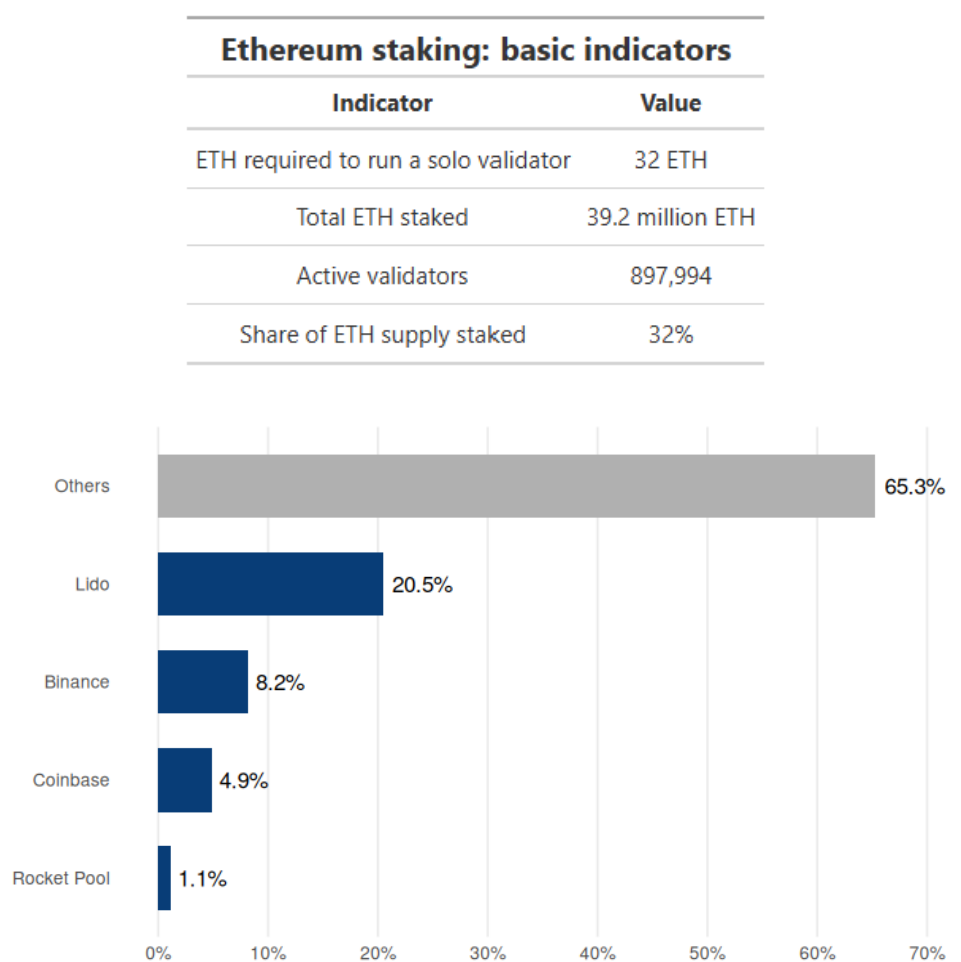
Concentration may also arise at the access layer, through exchanges, custodians, wallets and user interfaces. Many users do not run their own node or hold their own private keys, but instead access digital assets through centralised exchanges, custody services or wallet applications. Exchanges can decide which assets to list, how custody is organised, how compliance requirements are implemented and when trading is suspended. These intermediaries therefore create both dependencies and regulatory entry points.

User interfaces are particularly important in this respect. A smart contract may be directly accessible on-chain, but most users interact with it through a website or application. The front end can guide user choices, present or obscure information about risk, and restrict access to particular users. **Where the interface becomes the main practical route to a decentralised service, it can itself become an important point for regulation and consumer protection.**

These concentration points do not necessarily make the underlying protocol centralised, but they can determine where risk, dependency and regulatory leverage arise in practice. Where they involve identifiable intermediaries or infrastructure providers, existing regulatory and supervisory tools may be relevant, including requirements concerning operational resilience, outsourcing, governance, disclosure and business continuity. The relevant policy question is therefore not simply whether the protocol is decentralised, but which functions remain concentrated and which actors have the practical ability to influence them or reduce risk.

Ethereum illustrates this distinction because its base-layer architecture relies on distributed validation, but participation in validation is not evenly distributed. Figure 5 provides an overview of Ethereum staking and shows, as at 15 July 2026, the shares of staked ETH attributed to the four providers presented individually. The remaining share is grouped under “Others” and comprises staking activity not attributed to those four providers.

Figure 5: Ethereum staking and provider concentration (Panel A – Ethereum staking overview; Panel B – Shares of staked ETH by provider, 15 July 2026)



Sources: *Ethereum Foundation staking dashboard (staking overview)* and *Rated Network Explorer (staking-provider market shares, 15 July 2026)*.

Note: “Others” is a residual category calculated as 100% minus the combined shares of Lido, Binance, Coinbase and Rocket Pool. It is not an official provider category and includes the remaining staking providers and validators represented in the dataset, including smaller liquid-staking protocols, other exchanges, institutional staking providers, staking pools and independent validators.

For policymakers, decentralisation should be assessed layer by layer. The relevant question is which parts of the arrangement are distributed, which are concentrated and which actors have the practical ability to reduce risk.

A decentralised network can contain several layers in which economic or operational power becomes concentrated. Table 2 identifies the main concentration points and explains why they matter for regulation and resilience.

Table 2: Practical concentration points in decentralised networks

Layer	Main concentration point	Why it matters
Protocol (validation)	Large staking providers or pools	They may influence network security and resilience
Protocol (governance)	Core developers or governance processes	They may materially influence protocol changes and the evolution of the network
Access (entry points)	Major exchanges, custodians and wallet services	They are important entry and exit points for users
Access (infrastructure)	Node services, cloud providers and data tools	A failure can affect many applications at once
Access (interface)	Websites and applications used by ordinary users	They shape user choices and can show or obscure risk

Source: EGOV elaboration.

Table 3 brings together the two case studies while preserving the distinction between layers. It compares corporate stablecoin issuance at the instrument layer with a decentralised base-layer network at the protocol layer. The columns are not mutually exclusive: the same stablecoin may operate on the network described in the second column.

Table 3: Corporate stablecoin issuance and a decentralised base-layer network: two cases at different layers

Issue	Corporate stablecoin issuance (instrument layer)	Decentralised base-layer network (protocol layer)
Primary function	Issues a token intended to maintain a stable value and manages the reserve and redemption arrangement.	Records and validates transactions and provides infrastructure for assets and applications.
Main trust and dependency	Issuer, reserve assets, banks, custodians and legal rights.	Protocol rules, validators, software implementation, governance processes and infrastructure providers.
Main points of control	Issuance and redemption, reserve management, administrative keys and blacklisting powers.	Software upgrades, validation, staking concentration, client implementation and key infrastructure.
Main regulatory entry points	Issuer and CASPs, where they are within effective legal reach.	Identifiable governance actors and service or infrastructure providers, depending on their function and legal reach.
Typical risks	Reserve or redemption failure, issuer governance, cross-border legal reach and off-chain operational dependencies.	Software failures, governance or validation concentration, infrastructure dependency and diffuse accountability.
Key policy question	Who owes the promise, and can it be met under stress?	Who can change or materially influence the protocol, and how do users access it?

Source: EGOV elaboration.

Note: The two columns describe different layers and may form part of the same digital-asset arrangement.

5. AML and enforcement across issuer, protocol and access layers

AML rules show why the layers must be considered separately. **A corporate stablecoin may have an issuer able to implement controls, while the public network on which it circulates may permit peer-to-peer transfers without that issuer identifying every user.** In practice, customer due diligence and suspicious-transaction reporting are usually imposed on issuers and intermediaries, whereas **a public ledger mainly provides transaction traceability, which is valuable, but it does not by itself reveal the person behind an address.** Enforcement therefore depends on the combination of on-chain data, regulated access points and cross-border cooperation.

At EU level, MiCAR operates together with the EU AML/CFT framework. CASPs are obliged entities under the [Anti-Money Laundering Regulation](#)⁶ (AMLR) and must apply customer due diligence, transaction monitoring and suspicious transaction reporting requirements. The [Transfer of Funds Regulation](#)⁷ (TFR) also requires information on the originator and beneficiary to accompany crypto-asset transfers when a CASP is involved. **However, direct peer-to-peer transfers without a regulated intermediary remain more difficult to monitor.**

[FATF's 2025 update](#) notes progress in implementing AML/CFT standards for virtual assets and service providers, but also points to continuing **difficulties with licensing, registration, offshore providers and the identification of persons carrying out virtual asset service provider activities.** These difficulties concern identifiability and legal reach rather than centralisation alone. Requirements can be enforced more

⁶ Regulation (EU) 2024/1624 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing.

⁷ Regulation (EU) 2023/1113 on information accompanying transfers of funds and certain crypto-assets.

directly where the relevant actor is known and within jurisdiction, while offshore, hidden or highly distributed arrangements require other tools. Stablecoins create a special AML issue because they are useful for fast transfers of value and can move across borders, settle at any time and be used in many applications. **FATF has highlighted that most on-chain illicit activity now involves stablecoins**, according to the sources it cites in its 2025 update. This does not mean that stablecoins are mainly criminal tools, it means that criminals prefer instruments that are liquid, widely accepted and relatively stable. The same qualities that make stablecoins useful for lawful users can also attract illicit use. Going further, decentralised finance creates further difficulties, as **a decentralised exchange or lending protocol may operate through smart contracts rather than a traditional company, and if there is no identifiable operator, it is difficult to apply customer due diligence** in the usual way. Some projects may still have developers, governance token holders, front-end operators or fee recipients, while others may become more autonomous over time, so authorities need to look at the facts, and should seek whether there is a person or group that controls the service, profits from it, or can change its operation.

In AML enforcement, **the main challenge is to identify the points at which customer identification and transaction monitoring can operate effectively**. These points normally include stablecoin issuers, centralised exchanges, custodians, fiat on-ramps and off-ramps, and professional front-end operators. Where a small group controls a decentralised service or receives the main economic benefits from it, authorities should assess whether that group performs a regulated activity. International cooperation, blockchain analytics and stronger asset-recovery tools remain important where transactions cross borders or move outside regulated services.

6. Consumer protection and market integrity

EU law already addresses several of these risks, but the applicable framework depends on the legal nature of the asset and the service being provided. **MiCAR generally covers crypto-assets that do not fall under existing EU financial legislation**, subject to the exclusions and conditions set out in the Regulation. If a token qualifies as a financial instrument, it is excluded from MiCAR and is instead subject to [Markets in Financial Instruments Directive](#)⁸ (MiFID II), the [Market Abuse Regulation](#)⁹ (MAR) and other relevant securities rules. This distinction is relevant for consumers because two tokens may use similar technology while providing very different legal rights and protections.

Consumers often manage digital assets through simple applications and may not distinguish between a company-issued stablecoin, a market-priced crypto-asset used by a blockchain network and a tokenised security, because they may only see a price, a logo and a promise of easy access. **Since legal and technical differences are not always obvious to non-specialised users, there is a potential risk of misunderstanding, so regulation must make the information simple and comparable**. In the specific case of stablecoins, consumer protection should focus on the promise made by the issuer. Users need to know aspects such as whether the token can be redeemed, who can redeem it, how fast redemption should be, what fees apply, and what happens during stress; but they also need to know what assets support the token and whether there are limits or conditions. In short, **a stablecoin used as money should not be presented as equivalent in safety to a bank deposit or central bank money, since the applicable legal rights and protections are different**. MiCAR provides specific rights for holders of regulated stablecoins, as holders of ARTs have a permanent right of redemption, while holders of EMTs have a claim against the issuer and the right to redeem their tokens at any time and at par value. In addition, issuers must publish a crypto-asset white paper explaining the token, the issuer, the rights of holders and the main risks. Moreover, marketing communications must be fair, clear and not misleading. And finally, in the case of EMTs, the white paper must clearly state that the token is not covered by an investor compensation scheme or a deposit guarantee scheme.

⁸ Directive 2014/65/EU of the European Parliament and of the Council of 15 May 2014 on markets in financial instruments.

⁹ Regulation (EU) No 596/2014 of the European Parliament and of the Council of 16 April 2014 on market abuse.

In the case of decentralised networks, consumer protection is different. Here the risks may include price volatility, smart contract bugs, governance attacks, loss of private keys, scams, misleading interfaces and irreversible transactions. In addition, users need to understand that self-custody gives control, but also responsibility, because if a user holds the private key, this may have no easy remedy after a mistake, while if a user gives control to a custodian, the user bears an intermediary risk. MiCAR also places duties on authorised CASPs that provide custody, trading, exchange, advice or other services to clients. They must act honestly, fairly and professionally, provide fair and clear information, warn clients about risks, safeguard client assets, handle complaints and manage conflicts of interest. Furthermore, CASPs providing custody services have additional duties concerning custody agreements, records, security and responsibility for client assets. Besides, for certain direct purchases of crypto-assets other than ARTs and EMTs, MiCAR also gives retail holders a 14-day right of withdrawal, subject to specified conditions. Finally, DORA complements these protections by strengthening digital operational resilience requirements for financial entities in the crypto-asset ecosystem, including authorised CASPs and issuers of ARTs. It covers areas such as technology risk management, incident reporting, resilience testing and risks created by external technology providers.

Market integrity is also affected by governance, since centralised issuers and platforms can be required to prevent market abuse, manage conflicts of interest and disclose information, but if entities belonging to the same group issue a token, operate a trading venue, provide custody and run a lending product, conflicts of interest can become serious. MiCAR also establishes a specific market-abuse regime for crypto-assets admitted to trading or for which admission to trading has been requested. It prohibits insider dealing, the unlawful disclosure of inside information and market manipulation. Besides, persons professionally arranging or executing crypto-asset transactions must have systems to prevent and detect possible abuse and must report suspicious activity to the competent authority. [ESMA guidelines](#) support a common, risk-based and proportionate supervisory approach across the EU¹⁰. In this regard, the [ESRB](#) has identified multi-function groups in crypto-asset markets as a topic of macroprudential concern because scale and concentration may create important risks, showing this that the issue is not only the asset, but also the structure of the group around it. In decentralised markets, however, market abuse can take different forms: there can be manipulation through low-liquidity tokens, governance attacks, oracle manipulation¹¹, front-running or harmful use of transaction ordering. These problems are technical in form, but economic in substance, because they can damage users and confidence, so regulators may need new skills to understand them and to work with technology experts, academics, industry and other authorities. Education is useful, but it is not enough, given that users cannot be expected to understand every technical risk. Good regulation should reduce unnecessary complexity, require clear information and place duties on professional actors, and it should also prevent misleading marketing. For example, if a service is centralised, it should not be presented as fully decentralised; if a token depends on reserves, reserve information should be understandable; or if a user can lose all funds through a smart contract failure, this should be clearly explained before use. Applying these rules is more difficult where no identifiable issuer, operator or intermediary exists. Authorities should therefore assess the real allocation of control and determine whether an identifiable actor performs a regulated function, rather than relying on the project's description of itself as decentralised. A practical consumer protection test could be: **Can an ordinary user understand what they own, who they depend on, how they can exit, and what can go wrong?** If the answer is no, the product or service needs clearer disclosure or stronger controls. Finally, this test is useful across both models and keeps the focus on real user experience, not only on legal categories.

¹⁰ See ESMA, [Guidelines on supervisory practices to prevent and detect market abuse under the Markets in Crypto-Assets Regulation \(MiCA\)](#) (2025), and [Final Report on the Draft technical Standards specifying certain requirements in relation to the detection and prevention of market abuse under the Markets in Crypto Assets Regulation \(MiCA\)](#) (2024).

¹¹ Oracle manipulation is a cyberattack where a hacker manipulates a crypto-asset's price, tricking smart contracts into accepting false valuation data from an oracle (which is an on chain source for verified price data). This allows the attacker to exploit the system for unauthorised loans or fund withdrawals. See ESMA, [Crypto-assets and their risks for financial stability](#), Report on Trends, Risks and Vulnerabilities, Risk Analysis (2022).

7. Financial stability and monetary sovereignty

Financial stability concerns arise when digital assets become large, connected or difficult to replace. A small token used by a limited group of investors is unlikely, on its own, to create systemic risk, but **a large stablecoin used for trading, payments and collateral can create wider risk, given that if many users rely on it, a loss of confidence can spread quickly, and if the reserve assets are large, rapid sales can affect other markets.** In addition, if banks hold reserve deposits or provide services to issuers, **stress can move between crypto markets and traditional finance.** In particular, stablecoins are especially relevant because they aim to perform money-like functions and they may be used to store value, settle transactions or move funds across platforms. The [Bank for International Settlements](#) (BIS) has argued that stablecoins may be useful for some specific tokenisation activities, but they are not strong enough to become the main basis of the monetary system, especially when assessed against *singleness*, *elasticity* and *integrity*¹². In simple terms, private tokens may be useful in some contexts, but public money and regulated bank money remain central to trust in the wider economy.

Monetary sovereignty is linked to this issue because if most stablecoins are linked to the dollar, digital markets may become more dollar-based, and this can be relevant for countries or regions that desire payments and settlement to remain connected to their own currency. For the euro area, the concern is not only financial stability, but also whether the euro will have a strong role in digital finance. A market for euro-denominated stablecoins could support some European financial and payment activities, but it would still need strong safeguards, since weak rules could create new run risks or reduce trust. The ECB has repeatedly warned that stablecoins should be assessed by their functions, not only by their technology. Indeed, in a recent speech, the ECB's [President Christine Lagarde](#) noted that MiCAR brought stablecoins within the regulatory perimeter in 2024 to contain risks and protect financial integrity. This reflects a broader principle: **if a private token performs a money-like function, public authorities must understand the implications for payments, bank funding and monetary policy.**

Decentralised networks create different financial-stability questions. The crypto-asset used by such a network, such as ETH on Ethereum, is normally volatile and is not designed to maintain a fixed value. It may therefore be used differently from a stablecoin. However, **the network can support large markets, decentralised-finance applications and tokenised assets. If these activities become connected to banks, funds or market infrastructures, shocks may spread through leverage, liquidity mismatches, smart-contract failures or cyber incidents.**

Another highly relevant issue concerns tokenisation. When financial instruments are issued or traded on distributed ledgers, the line between crypto markets and traditional finance can become thinner, and this brings benefits, such as faster settlement and better automation, but also create new operational and legal risks. The [FSB](#) has noted that tokenisation can create run and spillover risks when backing assets or liquidity arrangements are weak. For legislators, tokenisation is therefore not only an innovation topic, but also a stability topic. The appropriate policy response is therefore not to treat digital assets as a homogeneous source of risk, but to map their interconnections carefully. Legislators consider how large the activity is, what assets support it, which banks or market infrastructures are involved, whether there is leverage, whether users can redeem quickly, and whether one failure could affect many services. This analysis needs to be done regularly as the market changes fast so a small product can become significant in a short time if it offers cheaper payments or if major platforms adopt it. The [limited initial use](#) of the EU DLT Pilot Regime illustrates a separate issue concerning tokenised market infrastructure rather than stablecoin issuance: legal permission must be accompanied by legal certainty, market demand, operational readiness and regulatory confidence.

Finally, **financial stability also requires clear exit arrangements, meaning users should know how they can leave a product;** issuers should plan for stress; supervisors should understand how reserves can be

¹² See BIS, [Annual Economic Report 2025](#), Chapter III: The next-generation monetary and financial system.

liquidated orderly; and for decentralised systems, authorities should know which service providers are most important for access and liquidity. Across the different layers, the relevant stress question is which function may fail, who has the capacity and responsibility to respond, and how losses or disruption could spread through the wider arrangement.

8. Concluding observations and key policy lessons

First, the analysis shows that centralisation and decentralisation are not mutually exclusive characteristics of a digital-asset arrangement, since they describe how particular functions are organised at different layers. A corporate stablecoin may combine centralised issuance, reserve management and redemption with transfer and validation on a distributed network. Conversely, a decentralised protocol may depend in practice on concentrated validation, infrastructure or user-access services. As a result, the relevant assessment is functional and layer-specific.

Second, neither centralisation nor decentralisation determines safety or supervisability on its own. What matters is whether control and dependencies are transparent, whether responsible actors or governance processes are within effective legal reach, and whether those actors or technical mechanisms can prevent, mitigate or remedy harm. Consequently, different layers may require different tools: issuer-based requirements where a legal promise exists; requirements addressing identifiable governance or control functions where legally applicable; and conduct, operational-resilience and AML/CFT obligations for service providers and gateways.

Third, MiCAR's differentiated treatment of ARTs, EMTs, other crypto-assets and CASPs is broadly consistent with this functional and actor-based logic. The analysis does not identify a general failure by MiCAR to differentiate among token types, **but main structural questions arise where no responsible actor can be identified, relevant actors are outside effective EU jurisdiction, or control becomes concentrated in protocol, infrastructure or access layers**; and separate implementation challenges concern supervisory coordination, data and technical capacity, and the interaction of MiCAR with DORA, the EU AML/CFT framework and other sectoral legislation.

Fourth, across the policy areas examined, the allocation of responsibility changes the available method of intervention rather than the underlying public objectives. Stablecoin oversight must focus on reserves, redemption and issuer governance; AML/CFT enforcement combines obligations on issuers and intermediaries with on-chain traceability; consumer protection depends on clear information and the regulation of access points; and financial-stability analysis must map concentration and interconnections across the full arrangement. Cross-border cooperation remains necessary wherever relevant functions, actors or dependencies are distributed across jurisdictions.

Ultimately, the policy response can be based on the location of the legal promise, the control of the protocol, and user access, translating the layered approach into three regulatory principles:

- Regulate the legal promise and the actor responsible for it at the instrument layer.
- Identify who can materially influence protocol governance and determine whether existing requirements or enforceable governance conditions can be applied.
- Regulate gateways and infrastructure providers according to their functions, and monitor concentration across all layers.

A related priority is to require accurate information about the real degree of decentralisation. Some projects are widely distributed and no single actor can easily change their rules, while other projects use decentralisation as a marketing claim while a small group of actors controls software upgrades, administrative keys, fees, voting or the main interface. Users and supervisors should be able to understand

who can change the system, who controls important keys, who receives income and which services the system depends on.

Concentration also needs to be monitored across the whole ecosystem. A stablecoin issuer can become a private payment gatekeeper, while a large exchange can become the main access point to many assets. A staking provider can control an important share of validation, while a small number of cloud or node providers can support many applications. Additionally, multi-function groups create an additional concern when the same group combines issuance, trading, custody, lending and other activities, because even though each service may appear manageable on its own, the group as a whole can create conflicts of interest, operational dependencies and possible channels of contagion.

Cross-border cooperation remains essential because issuers, reserves, service providers and users may be located in different jurisdictions. EU and third-country authorities therefore need effective information-sharing and consistent market-access conditions to reduce supervisory gaps and regulatory arbitrage.

Effective regulation also depends on supervisory capacity. Authorities need legal powers, but they also need suitable data and technical knowledge, because they must be able to assess smart contracts, wallet structures, reserve assets, validator concentration, custody arrangements, blockchain transactions, cyber risks and outsourcing. They should also understand how ordinary users interact with digital services, given that a protocol may be technically resilient while the website or custodian used by most consumers remains a weak point.

Proportionality remains important. A small experimental network should not be treated in the same way as a stablecoin used by millions of people for payments or settlement. Certainly, duties should be linked to the size, function, interconnectedness and risk of the activity but, at the same time, proportionality should not allow large firms or important gateways to avoid regulation by using complex structures or decentralised terminology.

The six control questions presented in Box 1 provide the starting point for this assessment, but legislators should then consider **four additional issues: whether the arrangement is used for payments, settlement, savings or collateral; how it connects with banks and financial markets; whether it can continue operating during a cyber or operational incident; and whether supervisors have sufficient data to understand how a failure could spread.** Together, these questions help apply existing rules to mixed business models that combine decentralised technology with centralised issuance, custody, access or governance.

The research question is therefore not whether centralised or decentralised digital-asset models should be preferred, but which characteristics determine where effective regulation and supervision can act. The analysis suggests that this requires identifying how each relevant function is organised, where control and dependencies arise, and which actors or mechanisms are capable of reducing risk. **Regulation should follow the legal promise, the control function and the user gateway, while supervision should assess how these elements interact across the arrangement as a whole.** Finally, this layered approach is consistent with MiCAR's differentiated treatment of issuers and CASPs, but it also explains why legal classification must be complemented by effective implementation, supervisory coordination, cross-border cooperation and regular monitoring of new concentration points.

GLOSSARY

Asset-referenced token (ART). A crypto-asset that seeks to maintain a stable value by referring to one or more currencies, assets, rights or a combination of them. ARTs are regulated under MiCAR in the EU.

Attestation. A report prepared by an independent professional to confirm that specified information, such as the amount and composition of stablecoin reserves, is presented correctly. It is more limited than a full financial audit.

Blockchain. A shared digital record in which transactions are grouped and stored in a way that makes earlier records difficult to change.

Blockchain analytics. The use of specialised tools to study transactions recorded on a blockchain and trace movements between addresses.

Centralised exchange. A company-operated platform on which users buy, sell or exchange crypto-assets. The platform normally controls access, trading rules and, in many cases, client assets.

Centralised model. An arrangement in which one identifiable organisation has decisive control over important functions, such as issuance, reserves, access or changes to the system.

Collateral. An asset provided to support a loan or another financial obligation. It may be sold if the borrower does not meet the obligation.

Corporate stablecoin. A stablecoin issued and managed by an identifiable private company that controls issuance, reserve management and redemption.

Crypto-asset. A digital representation of value or rights that can be transferred and stored electronically using DLT or similar technology.

Crypto-asset service provider (CASP). A professional firm providing services such as custody, trading, exchange, execution of orders, advice or transfers involving crypto-assets.

Custodian. A professional entity that holds or controls assets, or the keys needed to access them, on behalf of a client.

Custody. The safekeeping or control of assets or private keys on behalf of another person.

Decentralised autonomous organisation. An organisation or project whose decisions are partly made through blockchain-based rules and voting, rather than through a traditional company structure.

Decentralised exchange. A blockchain-based service that allows users to exchange crypto-assets through smart contracts, normally without a traditional central trading operator.

Decentralised finance (DeFi). Financial services, such as trading, lending or borrowing, provided through blockchain applications and smart contracts.

Decentralised network. A system in which transaction validation and network operation are distributed among many participants rather than controlled by one entity.

Distributed ledger technology (DLT). Technology that allows several participants to hold and update copies of the same digital record.

E-money token (EMT). A crypto-asset that seeks to maintain a stable value by referring to one official currency. EMTs are regulated under MiCAR in the EU.

Fiat currency. Official currency issued or recognised by a public authority, such as the euro or the US dollar.

Financial stability. A situation in which the financial system can continue to provide important services and absorb shocks without serious disruption.

Front end. The website, application or other interface through which an ordinary user accesses a blockchain service or smart contract.

Front-running. The practice of using advance knowledge of a pending transaction to place another transaction first and obtain an advantage.

Fungible. Interchangeable with another unit of the same type and value. One euro is normally fungible with another euro.

Gateway. A service that connects users to a blockchain or digital asset market, such as an exchange, custodian, wallet provider or front-end operator.

Governance attack. An attempt to gain enough influence over a blockchain project or voting system to approve changes that benefit the attacker or harm other users.

Governance token. A token that gives its holder the right to vote on certain decisions concerning a protocol or decentralised application.

Issuer. The person or organisation that creates a token and is legally responsible for the promise or rights attached to it.

Leverage. The use of borrowed funds or other financial techniques to increase exposure to an asset. It can increase both gains and losses.

Liquidity. The ability to buy, sell or redeem an asset quickly without causing a large change in its price.

Liquidity mismatch. A situation in which users can demand repayment quickly, while the assets used to meet those demands cannot be sold equally quickly or without losses.

Market abuse. Conduct that harms the fair functioning of a market, including insider dealing, unlawful disclosure of inside information and market manipulation.

Market capitalisation. The total market value of all units of an asset in circulation.

Market integrity. The degree to which a market operates in a fair, orderly and transparent way and is not distorted by manipulation or conflicts of interest.

Monetary sovereignty. The ability of public authorities to maintain control over their currency, monetary policy and the main monetary functions of the economy.

Multi-function group. A corporate group that performs several related activities, such as issuance, trading, custody and lending. This can create concentration and conflicts of interest.

Native asset. The crypto-asset that forms part of a blockchain's own design and is used to support its operation. Ether is the native asset of Ethereum.

Node. A computer running the software of a blockchain network and keeping or checking information about network transactions.

Off-chain. Taking place outside the blockchain. Stablecoin reserve assets held in banks or securities accounts are an example.

Off-ramp. A service that allows users to exchange crypto-assets for traditional money.

On-chain. Recorded or carried out directly on a blockchain.

On-ramp. A service that allows users to exchange traditional money for crypto-assets.

Open-source software. Software whose code can be viewed, checked, used or modified by the public under specified conditions.

Operational resilience. The ability of an organisation or system to continue providing critical services during technology failures, cyberattacks or other disruptions.

Oracle. A service that supplies a smart contract with external information, such as an asset price or an exchange rate.

Oracle manipulation. An attack in which false or distorted external data are supplied to a smart contract, causing it to make incorrect transactions or valuations.

Par value. The official value at which a token is issued or redeemed. An EMT linked to the euro should normally be issued and redeemed at one euro for each token.

Private key. A secret digital code that allows a user to control and transfer crypto-assets. A person who loses the key may lose access to the assets.

Proof-of-stake. A method used by some blockchain networks to validate transactions. Participants lock assets as a stake and can be penalised if they act dishonestly.

Protocol. The software rules that define how a blockchain network or decentralised application operates.

Pseudonymous. Identified by a digital address or code rather than directly by a person's real name. This does not mean that the activity is completely anonymous.

Redemption. The process through which a token holder returns a token and receives its reference value from the issuer.

Regulatory perimeter. The legal boundary that determines which products, services and actors are covered by a regulatory framework.

Reserve. Assets held by a stablecoin issuer to support the token's value and meet redemption requests. They may include cash, bank deposits or short-term (government) securities.

Run. A rapid and large number of redemption or withdrawal requests caused by a loss of confidence.

Self-custody. An arrangement in which a user controls their own private keys instead of relying on a professional custodian.

Settlement. The final transfer of an asset or payment that completes a transaction.

Single point of failure. One organisation, system or service whose failure can interrupt the operation of the wider arrangement.

Slashing. A penalty under proof-of-stake through which part of a validator's staked assets can be lost for dishonest or seriously incorrect behaviour.

Smart contract. Computer code stored on a blockchain that automatically performs actions when specified conditions are met.

Stablecoin. A crypto-asset designed to maintain a relatively stable value, normally by reference to a currency or other asset.

Staking. The locking of crypto-assets to participate directly or indirectly in the validation and security of a proof-of-stake network.

Staking pool or provider. A service that combines assets from several users or manages staking for them. Large providers can become concentration points.

Systemic risk. The risk that the failure of one important institution, market or service causes serious problems across the wider financial system.

Technology-neutral regulation. Regulation that focuses mainly on the activity and its risks rather than favouring or prohibiting a particular technology.

Tokenisation. The process of representing an asset, right or financial instrument through a digital token recorded on DLT.

Tokenised asset. An asset or right represented by a digital token. The legal rights normally depend on the asset behind the token and the applicable law.

Validator. A participant that checks transactions and helps create or confirm blocks in a proof-of-stake network.

Wallet. Software or hardware used to manage the private keys that give access to crypto-assets.

White paper. A disclosure document containing information about a crypto-asset, its issuer or offer, the rights attached to it, the technology used and the main risks.

LIST OF ABBREVIATIONS

AML	Anti-money laundering
AML/CFT	Anti-money laundering and countering the financing of terrorism
AMLR	Anti-Money Laundering Regulation
API	Application programming interface
ART	Asset-referenced token
BIS	Bank for International Settlements
CASP	Crypto-asset service provider
DeFi	Decentralised finance
DLT	Distributed ledger technology
DORA	Digital Operational Resilience Act
EBA	European Banking Authority
ECB	European Central Bank
EGOV	Economic Governance and EMU Scrutiny Unit
EMT	E-money token
ESMA	European Securities and Markets Authority
ESRB	European Systemic Risk Board
ETH	Ether (cryptocurrency)
EU	European Union
FATF	Financial Action Task Force
FSB	Financial Stability Board
MAR	Market Abuse Regulation
MiCAR	Markets in Crypto-Assets Regulation
MiFID II	Markets in Financial Instruments Directive
TFR	Transfer of Funds Regulation
US	United States
USD	United States dollar

USDC	USD Coin (stablecoin)
USDT	Tether (stablecoin)

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